

XRComm FCS 8T8R Developer Guide

Version 1.4.3 · June 2026

1. Building Your Own Application

The `XRComm_8T8R_hello_world/` folder contains a complete, working IQ consumer application. It demonstrates every step of the application lifecycle — registering with the platform, receiving per-channel IQ batches, computing power, reporting statistics, and shutting down cleanly.

Data Model The 8T8R hello-world example demonstrates processing independent streams from multiple configured channels across the 8 transmit and 8 receive channels.

Build the Hello-World Application

```
cd XRComm_8T8R_hello_world/  
mkdir build && cd build  
cmake .. -DXRCOMM_INCLUDE_DIR=../XRComm_platform_drivers/include  
cmake --build .
```

Run the Hello-World Application The platform driver must be running and IQ mode must be enabled before starting the application:

```
# Step 1 – enable IQ mode on the platform (from Client Computer):  
python3 XRComm_platform_gRPC/client/xrcomm_grpc_client.py set-ip on  
  
# Step 2 – run the application:  
./bin/hello_world
```

2. Application API Reference

Include `xrcomm_ip_client.h` to access the full API.

Function	Description
<code>xrcomm_ip_client_init_port</code>	Register for all channels of one HSP port as a combined stream.
<code>xrcomm_ip_client_wait_ready</code>	Block until the platform has opened this application's data ring.
<code>xrcomm_ip_client_running</code>	Returns true while the platform pipeline is running.
<code>xrcomm_ip_client_poll</code>	Return a pointer to the next IQ batch, or NULL if no data is available.
<code>xrcomm_ip_client_consume</code>	Release the batch returned by <code>poll()</code> . Must be called exactly once after each successful <code>poll()</code> .
<code>xrcomm_ip_client_get_dropped</code>	Return the number of batches the platform has dropped toward this application because its ring was full.

<code>xrcomm_ip_client_report_spectrum</code>	Publish per-application statistics visible in gRPC GetStats.
<code>xrcomm_ip_client_shutdown</code>	Deregister the application from the platform and release all resources.